Status	Finished
Started	Wednesday, 21 October 2026, 12:48 PM
Completed	Wednesday, 21 October 2026, 12:52 PM
Duration	4 mins 26 secs
Grade	**12.00** out of 15.00 (**80**%)

Question **1**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the first element in the list
b. Sort the list ✗
c. Divide the list into two equal parts
d. Compare the target element with the middle element in the list

Question **2**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The search ends successfully ✓
c. The search continues in the right sublist
d. The list is sorted

Question **3**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and unsorted ✓
b. When the list is small and sorted
c. When the list is large and unsorted
d. When the list is large and sorted

Question **4**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. n/2
c. log n
d. n ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Linear search ✓
b. Binary search
c. Both (1) & (3)
d. Hash search

Question **6**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500
b. 1000
c. 100
d. 10 ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues in the left or right sublist ✓
b. The search continues from the beginning
c. The list is sorted
d. The search stops

Question **8**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Jump search
c. Binary search
d. Linear search ✓

Question **9**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Insertion

b. Bubble ✓

c. Selection

d. Complexity

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It is slower than linear search for small lists
c. It requires the list to be sorted ✓
d. It does not work with negative numbers

Question **11**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all

b. When the item is the last element in the array

c. When the item is somewhere in the middle of the array ✓

d. Item is the last element in the array or item is not there at all

Question **12**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The middle element is chosen randomly

b. The search stops

c. The higher middle element is chosen as the middle element

d. The lower middle element is chosen as the middle element ✗

Question **13**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94

b. 90 and 99

c. 90 and 94 ✗

d. 89 and 99

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element

b. Sorting the list before searching

c. Checking each element sequentially ✓

d. Dividing the list in half repeatedly

Question **15**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Linear search
c. Sequential search
d. Binary search ✓